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Education

B.A. Media and Communications, Asbury University, May 2010.

Ph.D. Computer Science, University of Kentucky, April 2024.

"Flexible Attenuation Fields: Tomographic Reconstruction From Heterogeneous Datasets," Advisor: W. Brent Seales

Employment

- 2024–present Assistant Research Professor, Department of Computer Science, University of Kentucky, Lexington, KY.
- 2014–2024 Researcher and Project Manager, Department of Computer Science, University of Kentucky, Lexington, KY.
- 2011–2014 Video Production Coordinator, Center for Visualization and Virtual Environments, University of Kentucky, Lexington, KY.

Publications

Journal Articles

- Gokmen, Mahmut Selman, Caner Ozcan, Moneera N Haque, Steve W Leung, **Seth Parker**, Brent Seales, and Cody Bumgardner. "DINO-LG: Enhancing vision transformers with label guidance for coronary artery calcium detection". In: *Medical & Biological Engineering & Computing* (2026), pp. 1–18. DOI: 10.1007/s11517-026-03523-1.
- Nicolardi, Federica, Stephen Parsons, Daniel Delattre, Gianluca Del Mastro, Robert L. Fowler, Richard Janko, Tobias Reinhardt, **C. Seth Parker**, Christy Chapman, and W. Brent Seales. "Revealing Text from a Still-rolled Herculanum Papyrus Scroll (PHerc.Paris. 4)". In: *Zeitschrift für Papyrologie und Epigraphik* 229 (2024), pp. 1–13. DOI: 11588/959427.
- Dilley, Paul C, Christy Chapman, **C Seth Parker**, and W Brent Seales. "The X-Ray Micro-CT of a Full Parchment Codex to Recover Hidden Text: Morgan Library M.910, an Early Coptic Acts of the Apostles Manuscript". In: *Manuscript Studies: A Journal of the Schoenberg Institute for Manuscript Studies* 7.1 (2022), pp. 162–174.
- Chapman, Christy Y., **C. Seth Parker**, Ali Bertelsman, Kristina Gessel, Hannah Hatch, Kyra Seevers, James H. Brusuelas, Stephen Parsons, and W. Brent Seales. "The Digital Compilation and Restoration of Herculanum Fragment PHerc.118". In: *Manuscript Studies: A Journal of the Schoenberg Institute for Manuscript Studies* 6.1 (2021), pp. 1–32. DOI: 10.1353/mns.2021.0000.
- Parker, Clifford Seth**, Stephen Parsons, Jack Bandy, Christy Chapman, Frederik Coppens, and William Brent Seales. "From invisibility to readability: Recovering the ink of Herculanum". In: *PLOS ONE* 14.5 (May 2019), pp. 1–17. DOI: 10.1371/journal.pone.0215775.

- Parsons, Stephen, **C. Seth Parker**, and W. Brent Seales. "The St. Chad Gospels: Diachronic Manuscript Registration and Visualization". In: *Manuscript Studies: A Journal of the Schoenberg Institute for Manuscript Studies* 2.2 (2017), pp. 483–498. DOI: 10.1353/mns.2017.0022.
- Seales, William Brent, **Clifford Seth Parker**, Michael Segal, Emanuel Tov, Pnina Shor, and Yosef Porath. "From damage to discovery via virtual unwrapping: Reading the scroll from En-Gedi". In: *Science Advances* 2.9 (2016). DOI: 10.1126/sciadv.1601247.
- Segal, Michael, Emanuel Tov, William Brent Seales, **Clifford Seth Parker**, Pnina Shor, and Yosef Porath. "An Early Leviticus Scroll from En-Gedi: Preliminary Publication". In: *Textus* 26.1 (2016), pp. 29–58. DOI: 10.1163/2589255X-02601004.

Conference Proceedings

- Gokmen, Mahmut S., Moneera N. Haque, Steve W. Leung, Caroline N. Leach, **Seth Parker**, Stephen B. Hobbs, Vincent L. Sorrell, W. Brent Seales, and V. K. Cody Bumgardner. "A Framework for Cross-Domain Generalization in Coronary Artery Calcium Scoring Across Gated and Non-Gated Computed Tomography". In: *The Proceedings of the 2026 Amplify Informatics Conference*. American Medical Informatics Association, 2026. Forthcoming.
- Bhattacharyya, Ankan, **C. Seth Parker**, and W. Brent Seales. "Multispectral Imaging of Damaged Sacramental Journal Pages: A Preliminary Study". In: *Proceedings of 4th International Conference on Frontiers in Computing and Systems*. Ed. by Dipak Kumar Koley, Shubhajit Roy Chowdhury, Subhadip Basu, Dariusz Plewczynski, and Debotosh Bhattacharjee. Singapore: Springer Nature Singapore, 2024, pp. 659–679. ISBN: 978-981-97-2614-1. DOI: 10.1007/978-981-97-2614-1_47.
- Chapman, Christy, **Seth Parker**, Stephen Parsons, and W. Brent Seales. "Using METS to Express Digital Provenance for Complex Digital Objects". In: *Metadata and Semantic Research*. Ed. by Emmanouel Garoufallou and María-Antonia Ovalle-Perandones. Cham: Springer International Publishing, Mar. 2021, pp. 143–154. ISBN: 978-3-030-71903-6. DOI: 10.1007/978-3-030-71903-6_15.
- Parsons, Stephen, Jacob Chappell, **C. Seth Parker**, and W. Brent Seales. "Machine Learning Infrastructure on the Frontier of Virtual Unwrapping". In: *Proceedings of International Symposium on Grids & Clouds 2021 (ISCG2021)*. Academia Sinica Computing Centre (ASGC), Taipei, Taiwan (Online): Proceedings of Science, Mar. 2021, p. 15. DOI: 10.22323/1.378.0015.
- Gessel, Kristina, Stephen Parsons, **Clifford Parker**, and William Seales. "Towards Automating Volumetric Segmentation for Virtual Unwrapping". In: *Proceedings of the 25th International Conference on Cultural Heritage and New Technologies 2020*. Ed. by Wolfgang Börner, Hendrik Rohland, Christina Kral-Börner, and Lina Karner. Nov. 2020.
- Parsons, Stephen, Kristina Gessel, **Clifford Parker**, and William Seales. "Deep Learning for More Expressive Virtual Unwrapping". In: *Proceedings of the 25th International Conference on Cultural Heritage and New Technologies 2020*. Ed. by Wolfgang Börner, Hendrik Rohland, Christina Kral-Börner, and Lina Karner. Nov. 2020, pp. 203–207. DOI: 10.11588/propylaeum.1045.c14501.
- Ganio, Monica, Stephen Parsons, **Seth Parker**, Marie Svoboda, Brent Seales, and Catherine Schmidt Patterson. "Unbending light: new computational methods for the correction of 3D effects in scanning XRF". In: *Optics for Arts, Architecture, and Archaeology VII*. Conference Proceedings of SPIE Volume 11058. 2019. DOI: 10.1117/12.2525038.
- Parsons, Stephen, **C. Seth Parker**, Frederik Coppens, and W. Brent Seales. "Revealing "Invisible" Signals in CT with Machine Learning". In: *Bruker Micro-CT User Meeting*. Abstract Book. Mechelen, Belgium, June 2019, pp. 20–22.

- Parker, C. Seth** and W. Brent Seales. "Enhanced CT Analysis Using Volume Flattening". In: *Bruker Micro-CT User Meeting*. Abstract Book. Brussels, Belgium, June 2017, pp. 15–16.
- Parker, C. Seth**, W. Brent Seales, and Gregory Heyworth. "Reading the Invisible Library". In: *Bruker Micro-CT User Meeting*. Abstract Book. Mondorf-les-Bains, Luxembourg, May 2016, pp. 58–59.
- Parker, Clifford Seth**, William Brent Seales, and Pnina Shor. "Quantitative Distortion Analysis of Flattening Applied to the Scroll from En-Gedi". In: *Art & Archaeology, 2nd International Conference*. 2016.

Books and Anthologies

- Seales, W. Brent, **C. Seth Parker**, and Christy Chapman. "4.1.1.7 Virtual Unwrapping: A Computational Approach for Reading Damaged Manuscripts". In: *Textual History of the Bible*. Ed. by Armin Lange. Vol. 4. 2017. Chap. 1.1.7. DOI: 10.1163/2452-4107_thb_COM_225869.

Lectures, Presentations, and Preprints

- Parker, C. Seth**. "A Virtual Unwrapping Primer: The principles and practices for non-invasive, virtual text recovery" (Utah Valley University). Orem, Utah, Apr. 8–10, 2026. Invited speaker at *Artificial Intelligence and Ancient Civilization: A Conference on the Herculaneum Papyri*.
- Parker, C. Seth**. "To Herculaneum and Beyond: The Frontier Challenges of Virtual Unwrapping". Department of Computer Science Keeping Current Seminar (University of Kentucky). Lexington, KY, Feb. 4, 2026.
- Parker, C. Seth**, Una O'Leary, Michael J. Detisch, and W. Brent Seales. "Virtual unwrapping as a tool for the non-invasive, digital recovery of image content from brittle, photographic film rolls" (Palais des congrès de Montréal). Montreal, Quebec, Canada, Apr. 30–May 2, 2026. Poster presentation at *The 54th Annual Meeting of the American Institute for Conservation*.
- Parker, C. Seth**. "Multidisciplinary Research Breakout Session - Applications in AI. Panelist". NSF ACCESS Regional AI Workshop (Center for Computational Science, University of Kentucky). Lexington, KY, Oct. 8, 2025.
- Parker, C. Seth**. "The frontier challenges of virtually unwrapping unopened papyrus scrolls" (University of Cologne). Cologne, Germany, July 2025. Presentation at *The 31st International Congress of Papyrology*.
- Parker, C. Seth**. "Reading the invisible library. The digital restoration of hidden texts with virtual unwrapping and machine learning". Science Seminar (Taylor University). Upland, IN, Sept. 3, 2024.
- Parker, C. Seth**. "Scanning at scale. Digitizing the remaining closed scrolls". *Vesuvius Scrolls: From Ancient Manuscripts to Artificial Intelligence*. Mini-symposium (Michigan Institute for Data and AI in Society, University of Michigan). Ann Arbor, MI, Oct. 7, 2024.
- Parsons, Stephen, **C. Seth Parker**, Christy Chapman, Mami Hayashida, and W. Brent Seales. "Educelab-Scrolls: Verifiable Recovery of Text from Herculaneum Papyri using X-ray CT". Pre-print. 2024. DOI: 10.48550/arXiv.2304.02084.
- Parker, C. Seth**. "Metadata-enabled computational graphs". Department of Computer Science Keeping Current Seminar (University of Kentucky). Lexington, KY, Oct. 7, 2020.
- Parker, C. Seth**. "Reading the Invisible Library: Noninvasive Recovery of Text From Damaged Manuscripts". R-CHIVE Conference 2017 (University of Rochester and Rochester Institute of Technology). Rochester, NY, June 2017.

Projects and Software

Parker, C. Seth. *OpenABF. A single-header C++ library of angle-based flattening algorithms.* Comp. software. Jan. 2021. DOI: 10.5281/zenodo.4483858.

Parker, C. Seth, Kristina Gessel, Stephen Parsons, Jacob Chappell, Bruno Athie Teruel, Nikki Bentley, Ali Bertelsman, John Broadbent, Sydney Chapman, Abigail Coleman, Chao Du, Callie Gardella, Nick Graczyk, Hannah Hatch, Lula Hogg, Sean Karlage, Tam Nguyen, James Pack, David Pennington, Allison Revers, Mike Roup, Michael Royal, Kyra Seevers, Melissa Shankle, Raiffa Syamil, Ryan Taber, JP Posma, and Julian Schilliger. *Volume Cartographer. A cross-platform C++ library and toolkit for the recovery and restoration of damaged cultural artifacts.* Comp. software. Mar. 2021. DOI: 10.5281/zenodo.4604881.

Parker, C. Seth. *Structured Metadata Engine and Graph Objects Library.* Comp. software. Oct. 2020. DOI: 10.5281/zenodo.4134987.

Honors and Awards

Outstanding Student Paper Award, “From invisibility to readability: Recovering the ink of Herculaneum”, University of Kentucky, Department of Computer Science, April 2021

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